SPSS Lab Exercises

Student Survey Data Analysis

Applied Statistics · Student Survey Dataset · N = 42

This workbook contains three guided SPSS exercises based on the student survey dataset (N = 42). Each exercise states a research question, lists the relevant variables, provides numbered step-by-step SPSS instructions, and ends with a model solution and interpretation guide.

Contents at a Glance

#	Exercise	Method	Key Variables
1	Descriptive Statistics	Frequencies & Descriptives	life_satisfaction, gender, age, sleep_hours, financial_satisfaction
2	One-Way ANOVA	Between-groups comparison	life_satisfaction (DV), age_group (new IV)
3	Linear Regression	Simple linear regression	life_satisfaction (DV), gender_dummy (IV)

Exercise 1 — Descriptive Statistics

Background

Before running inferential tests, researchers always begin with descriptive statistics to understand the sample composition and the distribution of key variables. In this exercise you will produce frequency tables, summary statistics, and a histogram for the student survey data.

Research Questions

1. What is the gender distribution of the sample?
2. What are the mean, standard deviation, and range of **life satisfaction**?
3. How are **age**, **sleep hours**, and **financial satisfaction** distributed?
4. Are there any signs of skewness or outliers?

Variables

Variable	Role / Level	Description
gender	Nominal	Sex of student (Male / Female)
life_satisfaction	Scale	Overall life satisfaction, 0–10 scale
age	Scale	Age in years
sleep_hours	Scale	Average nightly sleep hours
financial_satisfaction	Scale	Satisfaction with finances, 0–10 scale

SPSS Instructions

Part A — Frequency table for Gender

1	**Analyze menu**	Click **Analyze** in the top menu bar.
2	**Frequencies**	Go to **Descriptive Statistics > Frequencies…**
3	**Select variable**	Move **gender** into the Variable(s) box.
4	**Add chart**	Click **Charts…**, choose **Bar charts** and select **Percentages**. Click Continue.
5	**Run**	Click **OK**. Inspect the frequency table and bar chart in the Output window.

Part B — Descriptives for scale variables

1	**Analyze menu**	Click **Analyze > Descriptive Statistics > Descriptives…**

2	**Select vars**	Move **life_satisfaction**, **age**, **sleep_hours**, and **financial_satisfaction** to the Variable(s) box.
3	**Options**	Click **Options…**. Tick: Mean, Std. deviation, Minimum, Maximum, Skewness, Kurtosis. Click Continue.
4	**Run**	Click **OK**. A Descriptive Statistics table appears in the Output window.

Part C — Histogram for Life Satisfaction

1	**Frequencies**	Click **Analyze > Descriptive Statistics > Frequencies…**
2	**Select var**	Move **life_satisfaction** into the Variable(s) box. Un-tick **Display frequency tables**.
3	**Charts**	Click **Charts…**, choose **Histograms** and tick **Show normal curve on histogram**. Click Continue.
4	**Run**	Click **OK** and inspect the shape of the distribution.

✔ Model Solution

Sample: 42 students — 24 females (57.1%) and 18 males (42.9%).

Life satisfaction: Mean = 6.67, SD = 1.87, range 2–10. The distribution is mildly negatively skewed, meaning most students report moderate-to-high satisfaction.

Age: Mean = 22.6 years, SD = 3.1, range 18–30. A predominantly young-adult sample.

Sleep hours: Mean ≈7.6 h, SD ≈0.9. Most students sleep within the recommended 6–9 h range.

Financial satisfaction: Mean = 5.83, SD = 2.64 — the wider spread reflects considerable variation in financial circumstances.

Tip: Skewness values between –1 and +1 indicate acceptable normality for subsequent parametric tests.

Exercise 2 — One-Way ANOVA — Life Satisfaction Across Age Groups

Background

One-Way ANOVA tests whether the means of a continuous dependent variable differ significantly across three or more independent groups. You will first create a new categorical variable (**age_group**) by recoding the continuous age variable, then test whether life satisfaction differs across the groups.

Research Question

Do students in different age groups (18–21, 22–25, 26–30) report significantly different levels of life satisfaction?

Hypotheses

H■: Mean life satisfaction is equal across all three age groups.

H■: At least one age group has a significantly different mean life satisfaction.

Variables

Variable	Role / Level	Description
life_satisfaction	Scale (DV)	Overall life satisfaction, 0–10
age	Scale (source)	Age in years — used to create age_group
age_group	Nominal (IV)	New variable: 1 = Young (18–21), 2 = Mid (22–25), 3 = Older (26–30)

SPSS Instructions

Part A — Create the age_group variable

1	**Transform**	Click **Transform > Recode into Different Variables…**
2	**Input variable**	Move **age** into the Input Variable → Output Variable box.
3	**Name output**	Type **age_group** as the Output Variable Name. Click **Change**.
4	**Recode values**	Click **Old and New Values…**. Use the Range option to set: 18–21 → 1, 22–25 → 2, 26–30 → 3. Click Add after each, then Continue.
5	**Run**	Click **OK**. The new variable age_group appears in Data View.

6	**Add labels**	In Variable View, click the Values cell for age_group and enter labels: 1 = 'Young (18-21)', 2 = 'Mid (22-25)', 3 = 'Older (26-30)'. Set Measure to Nominal.

Part B — Run the One-Way ANOVA

1	**Analyze menu**	Click **Analyze > Compare Means > One-Way ANOVA…**
2	**Variables**	Move **life_satisfaction** to Dependent List. Move **age_group** to Factor.
3	**Post Hoc tests**	Click **Post Hoc…**. Tick **Tukey** and **Games-Howell**. Set alpha = .05. Click Continue.
4	**Options**	Click **Options…**. Tick: Descriptive statistics, Homogeneity of variance test (Levene's), Means plot. Click Continue.
5	**Run & interpret**	Click **OK**. Review: (1) Descriptives table, (2) Levene's test, (3) ANOVA table, (4) Post Hoc comparisons.

✔ Model Solution

Levene's Test: If $p > .05$, equal variances assumed → use Tukey post hoc. If $p < .05$, variances differ → use Games-Howell.

ANOVA Table: Report as $F(df_{between}, df_{within})$ = [value], p = [value]. If $p < .05$ the null hypothesis is rejected.

Post Hoc: Tukey / Games-Howell comparisons identify which specific pairs differ. Report mean difference, 95% CI, and adjusted p-value for each significant pair.

Effect size: Compute eta-squared: $\eta^2 = SS_{between} / SS_{total}$. Values of .01, .06, .14 correspond to small, medium, large effects.

Write-up: 'A one-way ANOVA examined differences in life satisfaction across three age groups. Results indicated [significant/no significant] differences, F(2, 39) = X.XX, p = .XXX, η^2 = .XX. Post hoc tests revealed [describe any significant pairs].'

Tip: With ~14 participants per group, statistical power is limited. Always report effect sizes regardless of significance.

Exercise 3 — Simple Linear Regression — Predicting Life Satisfaction from Gender

Background

Simple linear regression estimates the linear relationship between one predictor and a continuous outcome. Because SPSS requires numeric predictors, nominal variables like gender must be **dummy coded** (0/1) before entry. You will recode gender and examine whether it predicts life satisfaction.

Research Question

Does gender predict life satisfaction among university students?

Hypotheses

H■: Gender does not significantly predict life satisfaction (β = 0).

H■: Gender significantly predicts life satisfaction (β ≠ 0).

Variables

Variable	Role / Level	Description
life_satisfaction	Scale (DV)	Overall life satisfaction, 0–10 (outcome)
gender	Nominal (source)	Male / Female — must be dummy coded first
gender_dummy	Scale (IV)	New variable: 0 = Male (reference), 1 = Female

SPSS Instructions

Part A — Dummy code gender

1	**Transform**	Click **Transform > Recode into Different Variables…**
2	**Input variable**	Move **gender** into the Input Variable → Output Variable box.
3	**Name output**	Type **gender_dummy** as the Output Variable Name. Click **Change**.
4	**Recode values**	Click **Old and New Values…**. Set: 'Male' → 0 and 'Female' → 1. Click Add after each, then Continue.
5	**Variable View**	In Variable View set Measure to Scale and add value labels (0 = Male, 1 = Female).

Part B — Check assumptions

1	**Normality of DV**	Click **Analyze > Descriptive Statistics > Explore…**. Move life_satisfaction to Dependent List. Under Plots tick Normality plots with tests and Histogram. Check the Shapiro-Wilk result ($p > .05$ supports normality).
2	**Group means**	Run **Analyze > Compare Means > Means…** to inspect mean life_satisfaction separately for males and females before fitting the model.

Part C — Run the regression

1	**Analyze menu**	Click **Analyze > Regression > Linear…**
2	**Dependent**	Move **life_satisfaction** to the Dependent box.
3	**Independent**	Move **gender_dummy** to the Independent(s) box. Method: Enter.
4	**Statistics**	Click **Statistics…**. Tick: Estimates, Confidence intervals (95%), Model fit, Descriptives. Click Continue.
5	**Residual plots**	Click **Plots…**. Set Y = *ZRESID and X = *ZPRED. Tick Normal probability plot. Click Continue.
6	**Run & interpret**	Click **OK**. Read: (1) Model Summary (R, R^2), (2) ANOVA table (F), (3) Coefficients (B, t, p, 95% CI).

✔ Model Solution

Group means check: Females M ≈6.5, Males M ≈6.9. The small raw difference already suggests a weak relationship.

Model Summary: R^2 indicates variance in life satisfaction explained by gender. Expect a low R^2 (< .05), indicating gender alone explains little variance.

ANOVA table: The F-test assesses overall model significance. If $p > .05$, gender does not significantly predict life satisfaction.

Coefficients table:
• Constant (B■): predicted life satisfaction for Males (gender_dummy = 0).
• gender_dummy (B■): change in predicted score when moving from Male to Female.
• If the 95% CI for B■ includes 0, the effect is not statistically significant.

Write-up: 'A simple linear regression examined whether gender predicts life satisfaction. Gender (0 = Male, 1 = Female) did not significantly predict life satisfaction, B = X.XX, t(40) = X.XX, p = .XXX, 95% CI [X.XX, X.XX], R^2 = .XX, F(1, 40) = X.XX, p = .XXX.'

Residual diagnostics: The Z-residual vs Z-predicted plot should show random scatter (homoscedasticity). The normal P-P plot should follow the diagonal line closely.

Tip: A non-significant result is a valid finding. Consider extending to multiple regression with additional predictors such as financial satisfaction or sleep hours.